Descriptive Statistics

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Relative & Absolute References

- Cell references can be relative or absolute. Using a dollar sign before a row and/or column label creates an absolute reference.
  - Relative references: A2, C5, D10
  - Absolute references: A2, $C5, D$10
- Using a $ sign before a row label (for example, B$4) keeps the reference fixed to row 4 but allows the column reference to change if the formula is copied to another cell.
- Using a $ sign before a column label (for example, $B4) keeps the reference to column B fixed but allows the row reference to change.
- Using a $ sign before both the row and column labels (for example, B4) keeps the reference to cell B4 fixed no matter where the formula is copied.

Functions

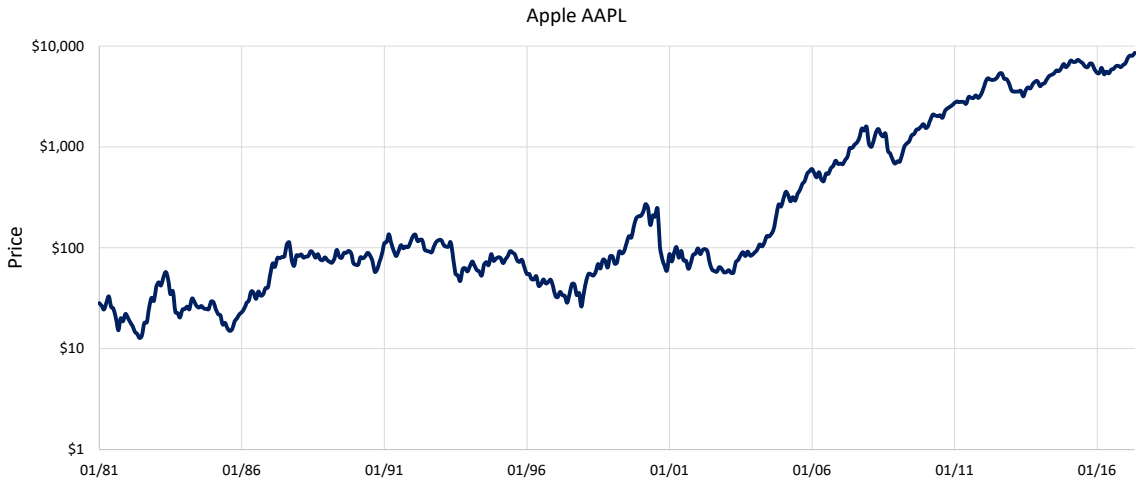
$= \min(data)$
$= \max(data)$
$= \text{sum}(data)$
$= \text{average}(data)$
$= \text{count}(data)$

Excel Add-Ins

- Microsoft Excel:
  - Analysis Toolpak
  - Solver
- Oracle provides:
  - Crystal Ball
  - See the software installation instructions on Sakai

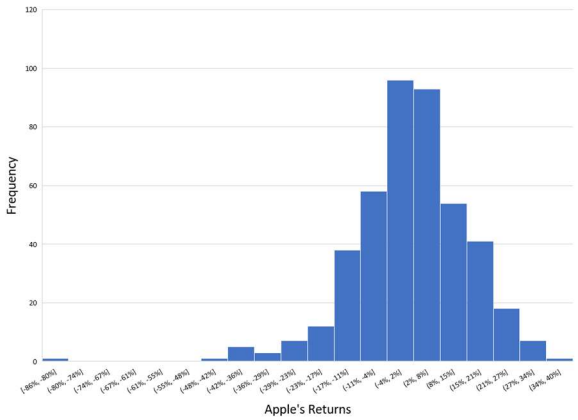
Line Charts

- Line charts display data over time



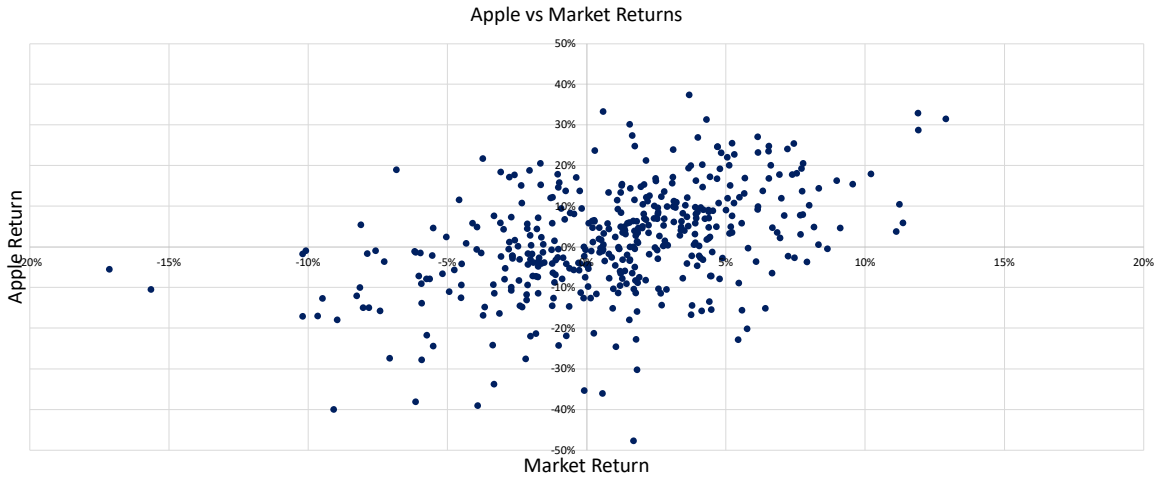
Histograms

- Plots the number of observations occurring in each group
  - Categorical variables naturally define the groups in a frequency distribution
  - For numerical data, we group data using bin ranges
  - Can use relative frequencies
  - Shows the distribution of a sample



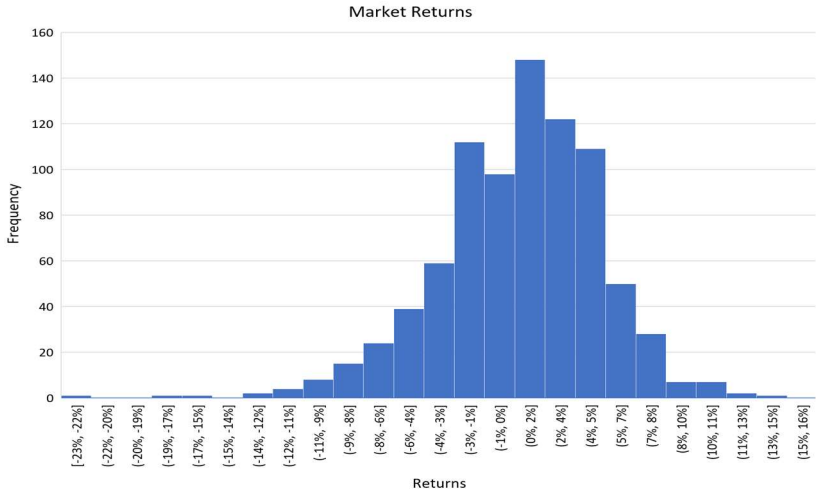
Scatter Charts

- Scatter charts show the relationship between two variables



Descriptive Statistics

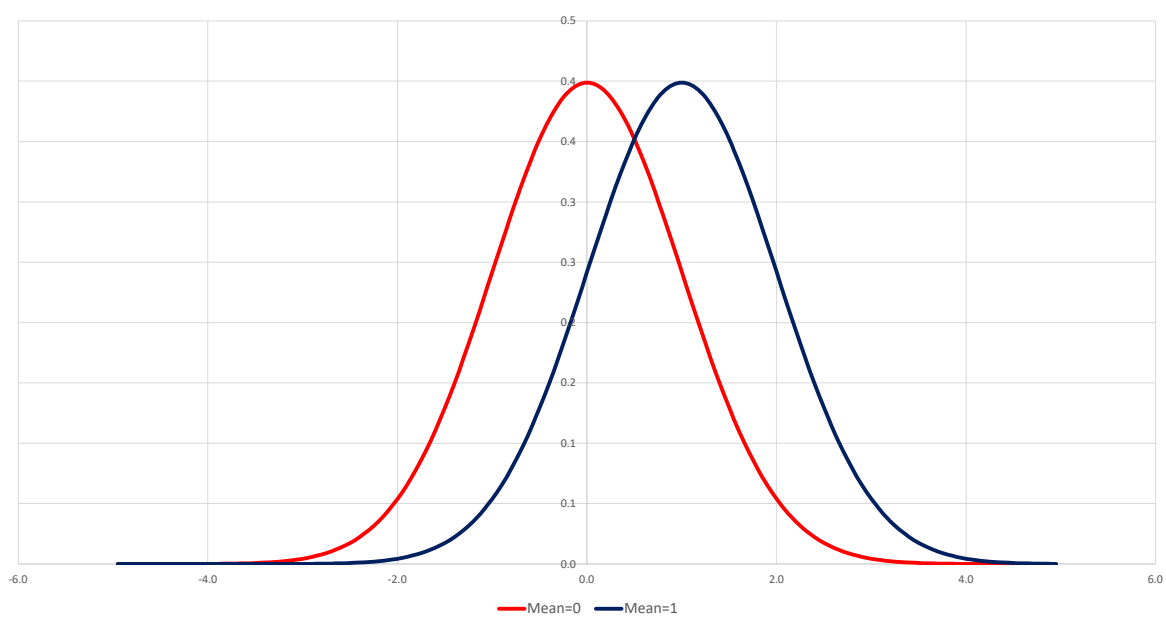
- Summarize a distribution's shape



Measure of Center: Mean

- Excel: =average(data)
- Also denoted as μ or $\bar{x}$
- Mean, average, expected value all synonymous

$$E[x] = \frac{1}{n}\sum_{i=1}^{n} x_i$$



Measures of Dispersion

- Dispersion refers to the degree of variation in the data, i.e. the spread of the distribution
- Important measures:
  - Variance σ^2
  - Standard deviation σ

Measures of Dispersion: Variance

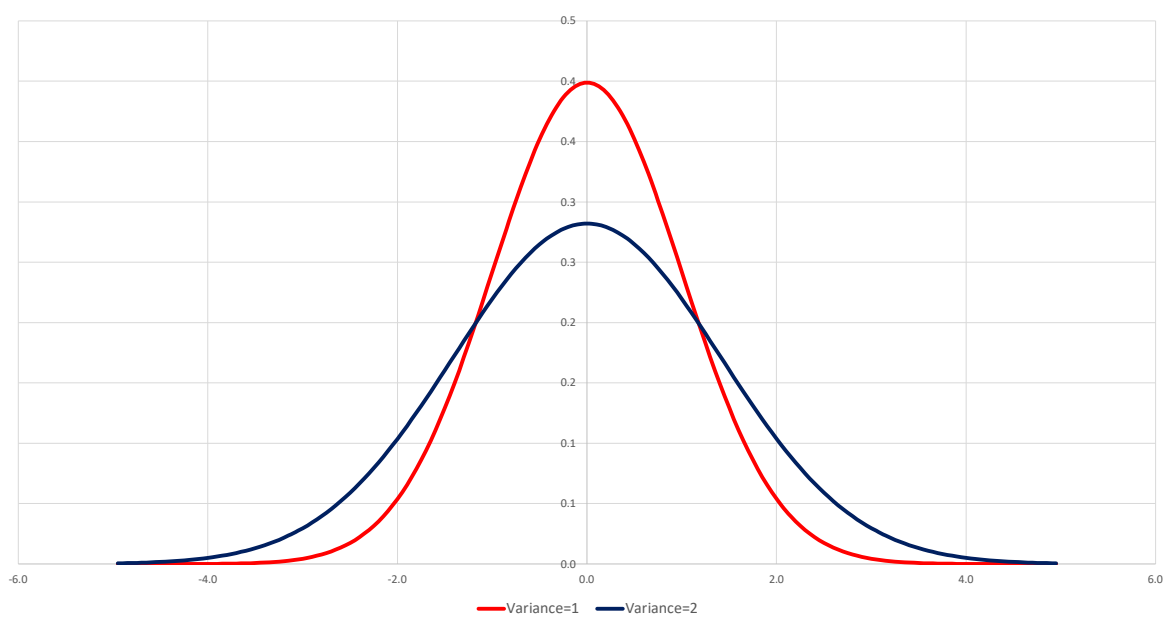
- Excel: =var.s(data)

$$Var(x) = \sigma_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Measures of Dispersion: Standard Deviation

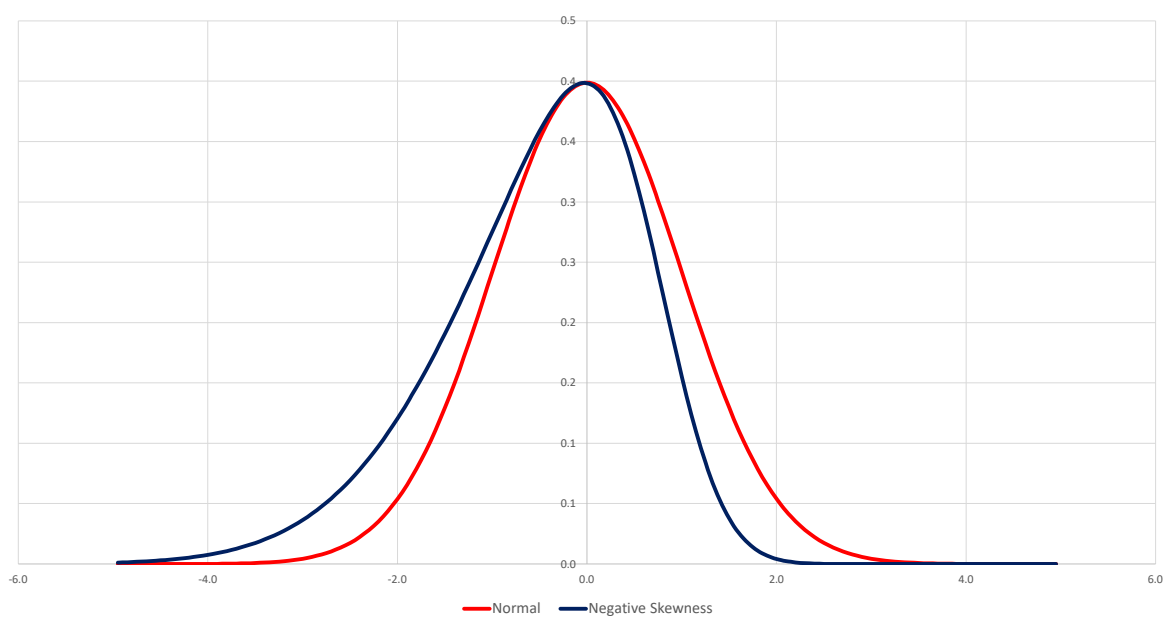
- The units of the variance is the square of the dimension of the observations, whereas the units of the standard deviation is the same as the data. This makes the standard deviation more practical to use in applications.
- $\sigma \geq 0$
- Excel: =stdev.s(data)

$$SD(X) = \sigma_x = \sqrt{\sigma_x^2}$$



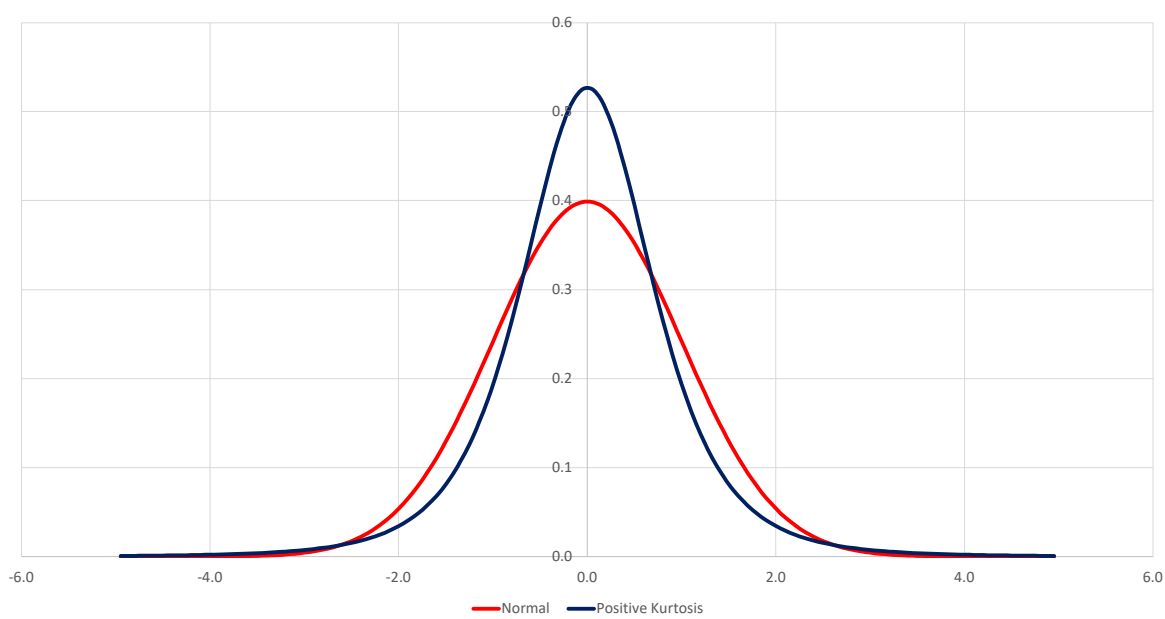
Measure of Asymmetry: Skewness

- Skewness describes the asymmetry of a distribution
  - Distributions that tail off to the right are positively skewed; those that tail off to the left are negatively skewed
  - Negative for left-skewed data
  - Positive for right-skewed data
  - The larger $|\mathrm{Skew}|$, the more asymmetry
- Excel function: =skew(data)



Measure of Peakedness: Kurtosis

- Kurtosis refers to the peakedness (i.e., high, narrow) or flatness (i.e., short, flat-topped) of a distribution
  - Kurtosis < 0 indicates the data is flat with a wide degree of dispersion
  - Kurtosis > 0 indicates the data is peaked and has fat tails
  - The larger $|\mathrm{Kurt}|$, the stronger the kurtosis
- Excel function: =kurt(data)



Measures of Association

- Two variables have a strong statistical relationship with one another if they appear to move together
- When two variables appear to be related, you might suspect a cause-and-effect relationship
- Sometimes, however, statistical relationships exist even though a change in one variable is not caused by a change in the other. Thus, correlation is not causation.

Measures of Association: Covariance

- Covariance is a measure of the linear association between two variables
- The covariance between X and Y is the average of the product of the deviations of each pair of observations from their respective means.
- Excel function: =covariance.s(data1,data2)

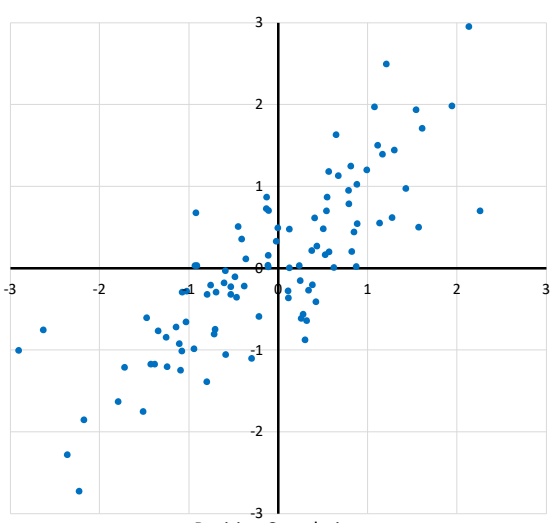
$$Cov(X,Y) = \sigma_{xy} = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$$

Measures of Association: Correlation

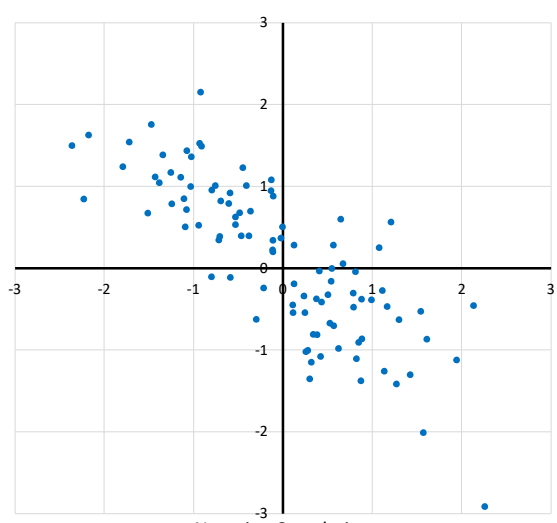
- Correlation is a measure of the linear relationship between two variables, which does not depend on the units of measurement
- The correlation coefficient is between -1 and 1
  - $r > 0$ indicates a positive relationship
    - When X goes up, Y goes up on average
  - $r < 0$ indicates a negative relationship
    - When X goes up, Y goes down on average
  - The larger $|r|$, the stronger the linear relationship
- Correlation does not imply causality
- Excel function: =correl(data1,data2)
- Also denoted r_{xy}

$$\rho_{xy} = \frac{Cov(X,Y)}{SD(X) \cdot SD(Y)} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

Examples of Correlation

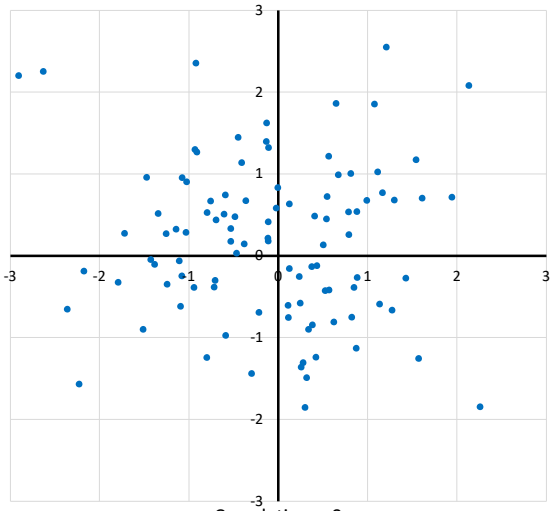


Positive Correlation

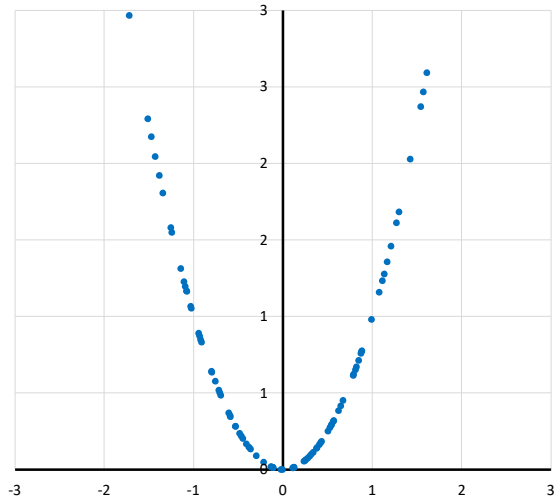


Negative Correlation

Examples of Correlation



Correlation = 0



Correlation = 0

Random Variables

- Random variable has random outcomes
- Outcomes have probabilities
- Two types of probability distributions
  - Discrete: outcomes are discrete values
  - Continuous: outcomes over continuous interval
  - Probability distributions sum to 1

Linear Combinations of Random Variables

Let a, b and c be constants and X, Y and Z be random variables

- $E[aX + bY] = aE[X] + bE[Y]$
- $Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2ab \cdot Cov(X, Y)$
- $Cov(aX + bY, cZ) = ac \cdot Cov(X, Z) + bc \cdot Cov(Y, Z)$

- Note: $Cov(X, X) \equiv Var(X)$

Expected Value

Let X be a discrete random variable with n outcomes

- X_i the i-th outcome
- $P(X_i)$ the probability of X_i

- Expected value

$$E[X] = \sum_{i=1}^{n} P(\mathrm{X}_i) \times \mathrm{X_i}$$

Variance & Standard Deviation

- Variance

$$Var(R) = \sigma^2 = \sum_{i=1}^{n} P(X_i) \times (X_i - E[X])^2$$

- Standard Deviation

$$SD(R) = \sigma = \sqrt{Var(R)} = \sqrt{\sigma^2}$$

Covariance & Correlation

- Covariance

$$Cov(X_1, X_2) = \sigma_{1,2} = \sum_{i=1}^{n} Prob_i \times \left(X_{1,i} - E[X_1]\right) \times \left(X_{2,i} - E[X_2]\right)$$

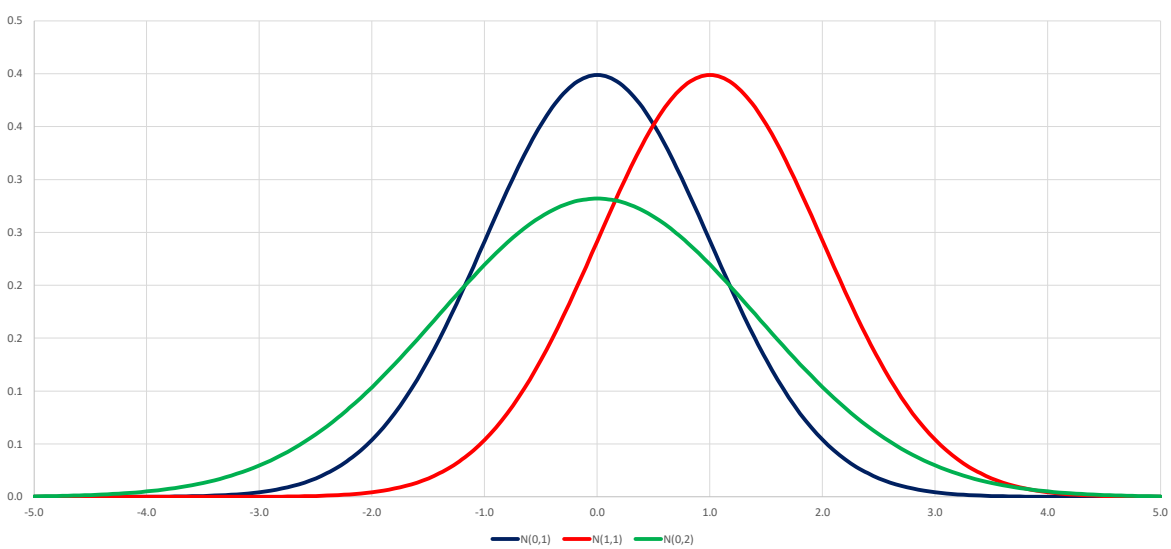
- Correlation

$$Corr(R_1, R_2) = \rho_{1,2} = \frac{\sigma_{1,2}}{\sigma_1 \sigma_2}$$

Normal Distribution

- Normal: $x \sim N(\mu, \sigma^2)$
  - Shape of distribution is bell curve
  - Mean of distribution is μ
  - Variance of distribution is σ^2
  - Symmetric: $\text{Skew} = 0$
  - $\text{Kurt} = 0$

Normal Distribution $x \sim N(\mu, \sigma^2)$



Normal Random Variable

- Probability of $-\infty \leq x \leq a$ is the area under curve from $-\infty$ to a

$$P(-\infty \leq x \leq a) = \text{norm.dist}(a, \mu, \sigma, \text{TRUE})$$

- Probability of a $\leq x \leq b$ is the area under curve from a to b

$$P(a \leq x \leq b) = P(-\infty \leq x \leq b) - P(-\infty \leq x \leq a)$$

- Probability of $a \leq x \leq \infty$ is the area under curve from a to ∞

$$P(a \leq x \leq \infty) = 1 - P(-\infty \leq x \leq a)$$

Normal Random Variable

- Probability of $-\infty \leq x \leq a$ is the area under curve from $-\infty$ to a

$$P(-\infty \leq x \leq a) = \text{norm.dist}(a, \mu, \sigma, \text{TRUE})$$

- The value on the x-axis, a, that corresponds to probability $p = P(-\infty \leq x \leq a)$

$$a = \text{norm.inv}(p, \mu, \sigma)$$